

SQL-Based Database Management System : airbnb_datamart

Phase 3 Finalisation Abstract | DLBDSPBDM01 | Kashish Paresh Gada | Matric. No. 102202657

Project Overview

This document summarises the finalised airbnb_datamart project, a relational database management system built in MySQL to simulate an Airbnb-style accommodation booking platform. The system covers the full booking lifecycle: user registration, property listing, reservation, payment, host payout, fee calculation, reviews and social messaging. Three user roles are modelled under the User entity: Hosts (list properties, manage availability and house rules), Guests (browse, book, pay and review) and users holding both roles simultaneously. The platform enforces a real-world fee structure: a 12% guest service fee and a 3% host service fee per booking.

Development Process Summary

Development followed a structured three-phase approach. The Conception phase produced a requirements specification and a relational Entity Relationship Model in crow's-foot notation covering 25 entities and 4 junction tables. The Development phase produced three validated SQL files: schema.sql (29 tables with all constraints), inserts.sql (minimum 20 rows per table, end-to-end business scenarios), and queries.sql (13 test-case queries). A combined complete_scenario.sql was also produced. All files were executed and verified on MySQL 8.0.46 with exit code 0 and zero errors.

Data Population Strategy

All 29 tables contain a minimum of 20 rows, totalling approximately 944 rows. Data was designed as coherent end-to-end scenarios: each of the 24 bookings has a matching payment, payout and two fee rows. The 18 completed bookings each have at least one review with per-category ratings. The 43 user records are split into 20 host-only, 20 guest-only and 3 dual-role users.

Constraint Implementation and Data Integrity

Data integrity is enforced at database level through four constraint types. PRIMARY KEY: every table has a named 'pk_' constraint with readable VARCHAR(50) IDs. FOREIGN KEY: all 37 ERD relationships enforced with appropriate ON UPDATE CASCADE / ON DELETE CASCADE or RESTRICT actions. UNIQUE: 16 constraints on identity fields and 1:1 relationship. CHECK: 19 business rules including ratings 1-5, check_out after check_in, non-negative amounts, valid lat/long ranges, and no self-reviews, self-messages or self-connections. InnoDB engine with MySQL 8.0.16+ is required for CHECK enforcement.

System Metadata

Property	Value
Database Name	airbnb_datamart

Total Tables	29 (25 entities + 4 junction tables)
Engine / Charset	InnoDB / utf8mb4
PRIMARY KEY constraints	29
FOREIGN KEY constraints	37
UNIQUE constraints	16
CHECK constraints	19
Total Rows (all tables)	~944
Total Database Size	1.11 MB
DBMS / Tool	MySQL 8.0.46 / MySQL Workbench 8.0

Row count per table:

Table	Rows	Table	Rows
country	20	city	24
address	30	user	43
host	23	guest	23
socialaccount	26	paymentmethod	26
payoutmethod	26	propertytype	24
roomtype	20	cancellationpolicy	20
amenity	25	reviewcategory	20
listing	24	photo	48
houserule	30	availabilitycalendar	48
listingamenity	72	booking	24
payment	24	payout	24
fee	48	review	30
reviewrating	90	message	30
wishlist	24	wishlistitem	48
userconnection	30		

Final Deliverables

- 01-Concept: Phase 1 PDF - requirements specification, ERD (crow's-foot, 29 tables), data dictionary
- 02-Development: Phase 2 presentation PDF (22 slides), SQL files (schema, inserts, queries, complete_scenario)
- 03-Finalisation: This abstract (PDF), README installation manual (.md + PDF)
- Combined PDF: single document with all content from all three phases including screenshots
- ZIP archive: Gada-Kashish_102202657_DLBDSPBDM01_Data_Mart_Submission_all.zip